Range Data Results

Sophie Litschwartz

2024-10-09

What is the distribution of the range of estimates across simulations when all the estimands are the same?

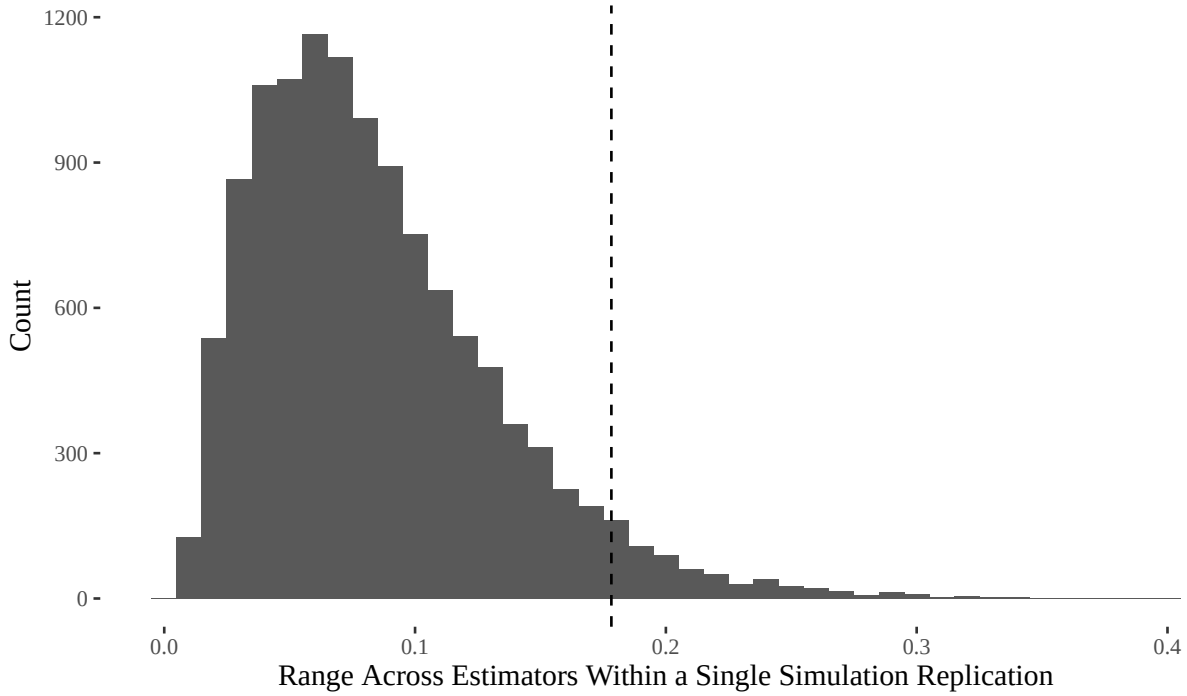
In this group of simulations all the estimands are equal because there is no treatment effect variation

Parameter Values for Simulations in This Histogram

K	J.bar	p.bar	sd.district	ICC.district	variable.J	variable.n	variable.p	n.bar	sd.school	ICC.school
10	10	0.5	0.1	0.2	TRUE	TRUE	FALSE	25	0	0.2

size.impact.cor.district	size.impact.cor.school	proptx.impact.cor.district	size.intercept.cor.district	cor.strength.school
0	0	0	0	0

This is a histogram of the magnitude of the ranges across the different estimators across the replications of a fixed data generating process. The dotted line is at the 95% percentile across the simulation replications.



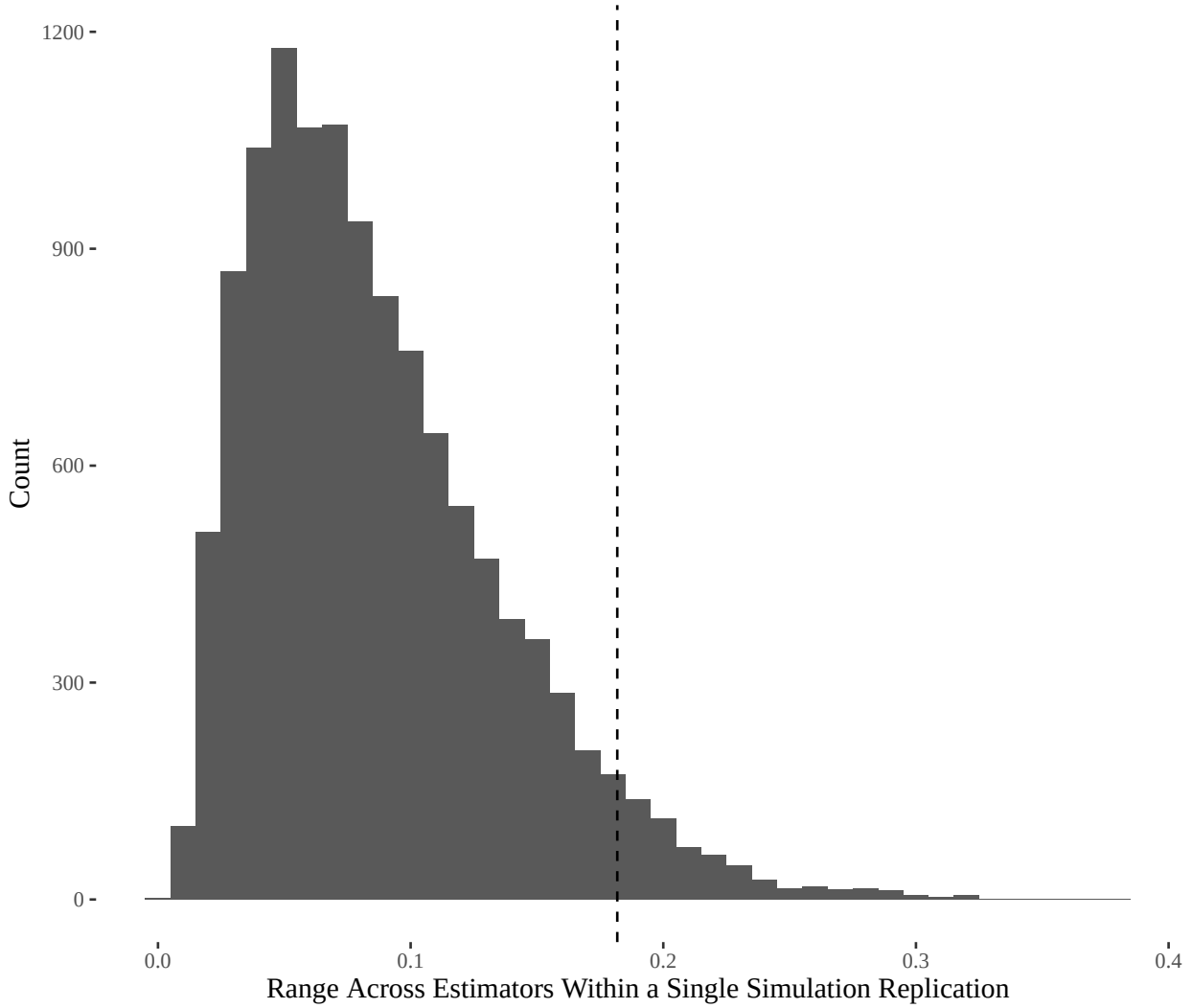
What is the distribution of the range of estimates across simulations when the estimands are different and the parameters for the data generating process roughly matches the values in the empirical papers?

Parameter Values for Simulations in This Histogram

K	J.bar	p.bar	sd.district	ICC.district	variable.J	variable.n	variable.p	n.bar	sd.school	ICC.school
10	10	0.5	0.1	0.2	TRUE	TRUE	FALSE	25	0.1	0.2

size.impact.cor.district	size.impact.cor.school	proptx.impact.cor.district	size.intercept.cor.district	cor.strength.school
0	1	0	0	0.1

This is a histogram of the magnitude of the ranges across the different estimators across the replications of a fixed data generating process. The dotted line is at the 95% percentile across the simulation replications.n




```
## `summarise()` has grouped output by 'ID'. You can override using the `.groups`
## argument.
```

How does the range across estimators vary as the average number of sites varies?

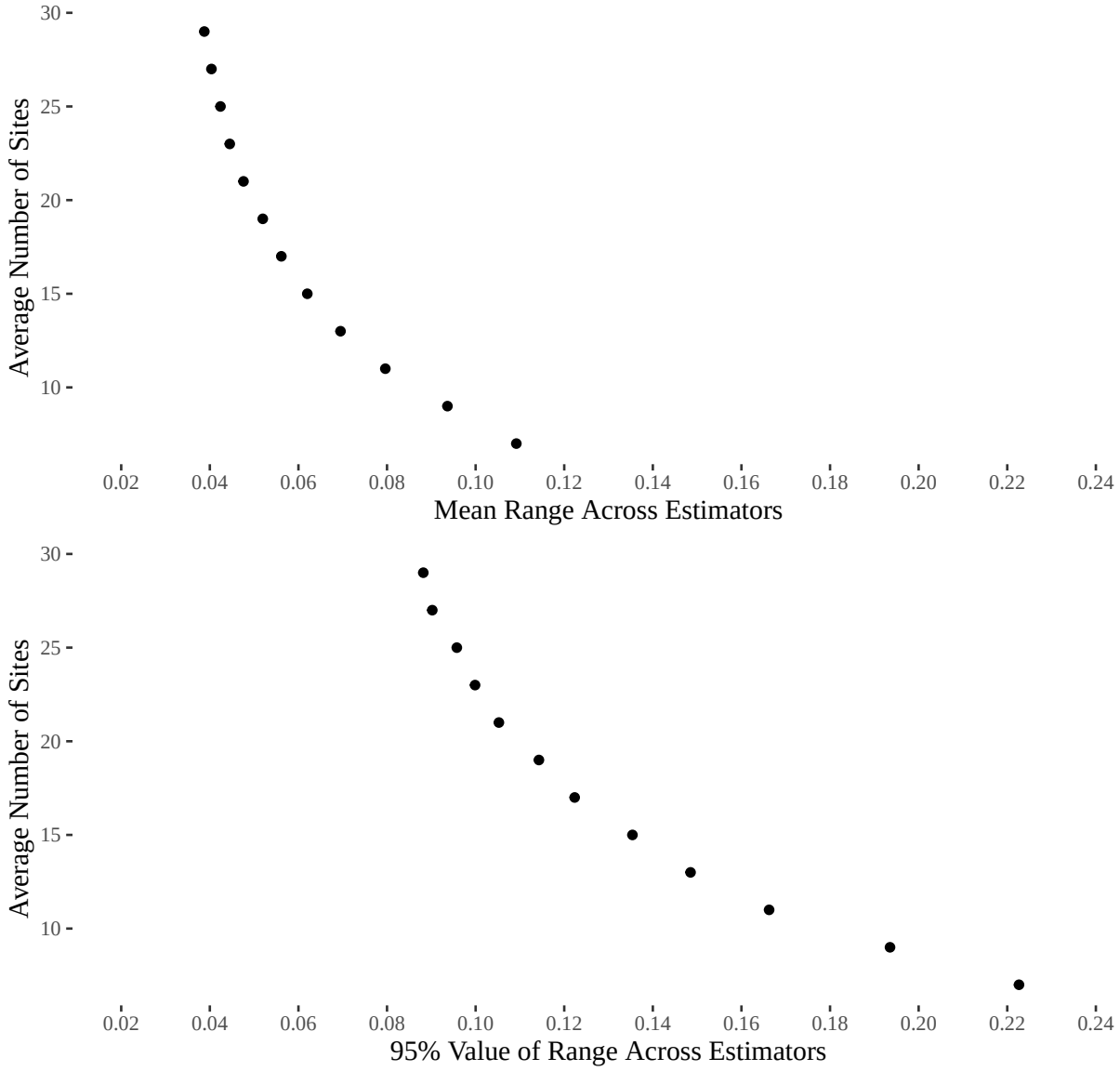
In these graphs the average number of sites varies as all other parameter values are kept at values designed to mimic the empirical data. These parameter values are as follows:

Parameter Values for Simulations in This Plot

K	p.bar	sd.district	ICC.district	variable.J	variable.n	variable.p	n.bar	sd.school	ICC.school
10	0.5	0.1	0.2	TRUE	TRUE	FALSE	25	0.1	0.2

size.impact.cor.district	size.impact.cor.school	proptx.impact.cor.district	size.intercept.cor.district	cor.strength.school
0.2	1	0	0	0.2

The range in values across estimators goes down as the number of sites goes up.



How does the range across estimators vary as the average site size varies?

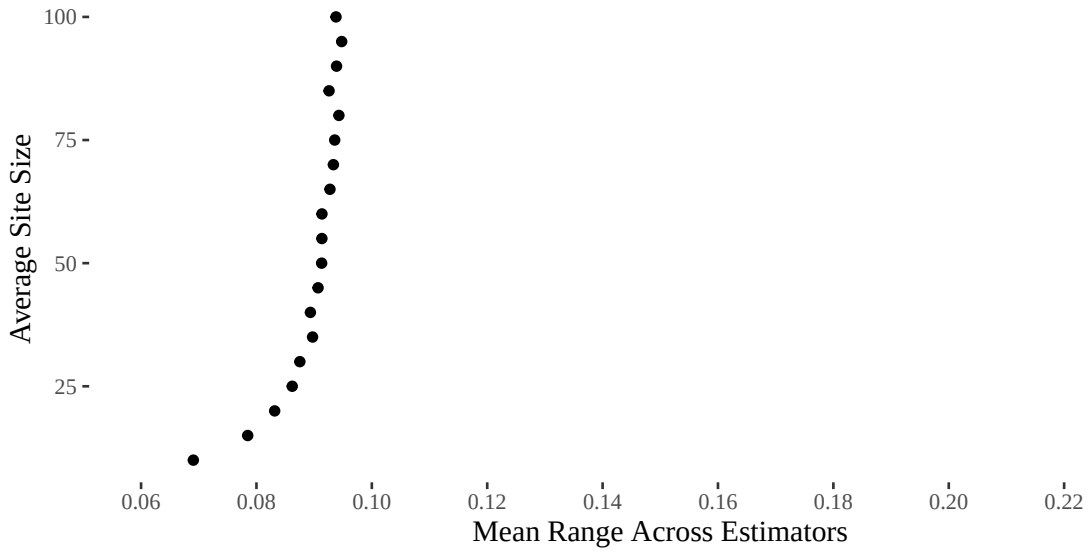
In these graphs the average site size varies as all other parameter values are kept at values designed to mimic the empirical data. These parameter values are as follows:

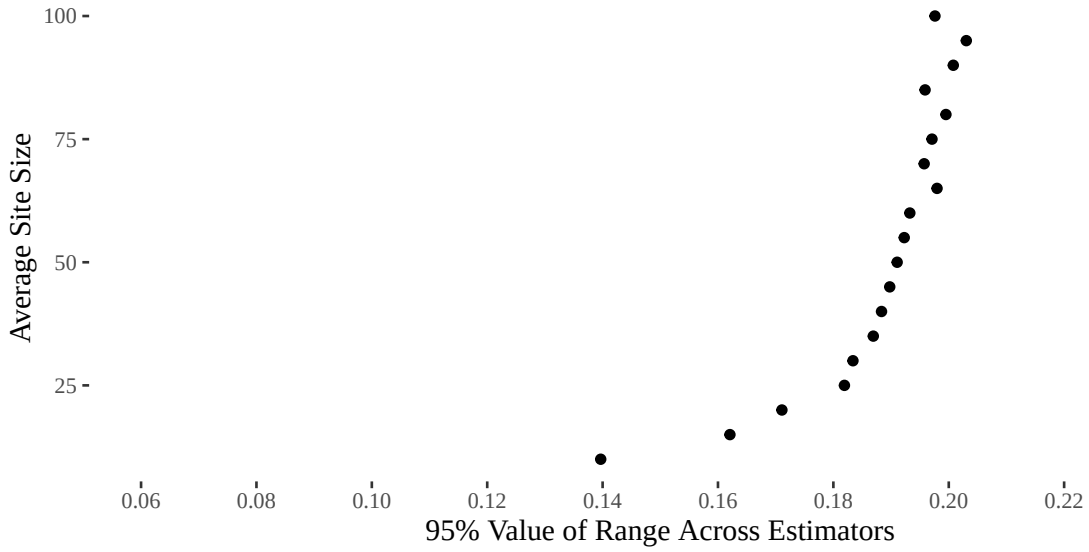
Parameter Values for Simulations in This Plot

K	J.bar	p.bar	sd.district	ICC.district	variable.J	variable.n	variable.p	sd.school	ICC.school
10	10	0.5	0.1	0.2	TRUE	TRUE	FALSE	0.1	0.2

size.impact.cor.district	size.impact.cor.school	proptx.impact.cor.district	size.intercept.cor.district	cor.strength.school
0.2	1	0	0	0.2

The range in values across estimators goes up as the site size goes up .





How does the range across estimators vary as the correlation between the site size and the treatment effect increases?

In these graphs the correlation between the site size and the treatment effect varies as all other parameter values are kept at values designed to mimic the empirical data. These parameter values are as follows:

Parameter Values for Simulations in This Plot

K	J.bar	p.bar	sd.district	ICC.district	variable.J	variable.n	variable.p	n.bar	sd.school
10	10	0.5	0.1	0.2	TRUE	TRUE	FALSE	25	0.1

ICC.school	size.impact.cor.district	proptx.impact.cor.district	size.intercept.cor.district
0.2	0	0	0

The range in values across estimators does not change much as the correlation between the site size and the treatment effect changes, implying that differences in the estimates is not heavily driven by differences in the estimands

